WRITING AND NOTEBOOK STYLE

The Okafor Lab's style guide for notebooks, figures, manuscripts and data. Version 1.4, January 2026. Adaeze Okafor, with Priya Raghunathan.

Good science is reproducible by someone who is not you, including you in three years' time. These conventions exist so that anyone in the lab can find your samples, understand your notebook and rebuild your figures from raw data. They are not optional. They apply to everyone, including rotation students and undergraduates.

1 THE ELECTRONIC LAB NOTEBOOK (ELN)

We use the Westbrook ELN. Each person has a notebook inside the "Okafor Lab" group; Adaeze and Marisol can read all of them. **Write the entry on the day you do the work.** Title each entry "YYYY-MM-DD – short description" (for example, "2026-10-13 – HSP70 western, HEK293 heat shock").

Every entry must contain:

1. The date and your name.
2. The aim: what question the experiment asks.
3. Sample IDs, and for cells the line and passage number.
4. Reagents, with **lot numbers** for antibodies, enzymes, serum and kits.
5. The protocol: a link to the SOP and its version, and **every deviation** from it.
6. The instruments and settings used (for ultracentrifuge runs, the rotor-log line as well).
7. Observations and results, *including failures*.
8. The path to the raw data on `okafor-nas`.
9. Conclusions and next steps.

Never delete content. If something is wrong, strike it through and add a dated note explaining why. Adaeze reads one randomly chosen entry per person each month and discusses it at your 1:1.

2 NAMING CONVENTIONS

Item	Pattern	Example
Sample ID	INITIALS-YYMMDD-NN	EP-261013-03
Data file	`YYYYMMDD_Initials_Experiment_Instrument_vNN.ext`	`20261013_EP_WB-HSP70_imager_v01.tif`
Cryovial label	Cell line, passage, date, initials, sample ID	HEK293 p12 2026-10-13 EP EP-261013-05
Plasmid	pOKA followed by four digits, from the plasmid database	pOKA0412

No spaces in file names. Dates are always in ISO order (year, month, day). Versions are always two digits (v01, v02 and so on). Never call a file "final". Every tube carries at least the sample ID; if it holds more than one sample, the box does.

3 Figures

- **Fonts:** Arial or Helvetica, 7–9 pt at final printed size. Panel labels are bold capitals (A, B, C).
- **Colour:** use the colour-blind-safe **Okabe–Ito palette** for categories: black #000000, orange #E69F00, sky blue #56B4E9, bluish green #009E73, yellow #F0E442, blue #0072B2, vermillion #D55E00, reddish purple #CC79A7. Use *viridis* or *cividis* for continuous scales. **Never use red–green contrasts or rainbow ("jet") colour maps.**
- **Show the individual data points.** Use dot plots, box plots with the points overlaid, or bars *with* points. Bar-only charts are not accepted in lab figures.
- **Error bars:** say in the legend what they show (SD, SEM or 95% CI). Our default for describing variability is SD.
- **Report n and say what it is.** A biological replicate is an independent culture, experiment or animal. Technical replicates are averaged and do not count towards n. Name the statistical test, give exact p-values ($p = 0.013$; report very small values as $p < 0.001$), and state any correction for multiple comparisons.
- **Blots:** show molecular-weight markers and a loading control from the same membrane, and keep the uncropped originals for the supplement.
- **Micrographs:** include scale bars. Compare only images acquired with the same settings. Apply linear adjustments only, and apply them to the whole image and equally to every image being compared.
- **Export:** plots as vector PDF or SVG; images as TIFF at 300 dpi or more. Keep the source (Prism file or R/Python script) in the project folder.

4 Manuscripts and other writing

- **Voice and tense:** active voice, first person plural ("We purified the complex…", not "The complex was purified…"). Past tense for your results, present tense for established knowledge.
- **Units:** SI, with a space between the number and the unit (5 mL, 37 °C, 10 min), except for percentages (10%). Give centrifugation as relative centrifugal force ($\times g$), never rpm. Write litres with a capital L (mL, µL).
- **Abbreviations:** define each at first use in the abstract and again in the main text, and keep them to a minimum.
- **References:** use the "Okafor Lab" Zotero group library.
- **Sending drafts to Adaeze:** email a Word file (.docx) with line numbers and tracked changes on, with the subject line "[DRAFT for AO] Title – date". Name the file Title_vNN_Initials.docx

and put the figures at the end. **Do not send PDFs**, and do not send Google Docs unless you have agreed on it in advance.

- **Turnaround.** Adaeze returns abstracts and conference materials within 3 working days, fellowship or grant sections within 1 week, and full manuscript drafts within 2 weeks. Send anything with a deadline **at least 2 weeks before it** (1 week for abstracts). Give co-authors 1 week to comment.
- Before submission, every co-author approves the final version and the data are deposited.

5 Data management

- **Raw data live on the lab server**, under `\\okafor-nas.westbrook.edu\raw\<initials>\<year>`. The raw folder is write-once: files become read-only after you copy them in. Do your analysis on copies in `\\okafor-nas\work\<initials>`.
- **Instrument PCs are not storage.** Copy your data off at the end of every session. Instrument PCs are wiped on the first Monday of each month.
- **Backups:** hourly snapshots on `okafor-nas`, a nightly backup to Westbrook Research Storage, and a weekly off-site copy.
- **Never** keep the only copy of anything on a laptop, a USB drive or a personal cloud account.
- **Cryo-EM data** stay on the facility's storage until Lucas Ferreira moves them to the lab's cryo-EM volume (`okafor-em`).
- **Retention:** raw data are kept for at least 7 years after publication. When you leave the lab, you go through the exit checklist with Marisol and Adaeze: raw data on the server, the ELN complete, and your freezer boxes labelled and in the inventory.